Team Diamond (DVA)

Business Insights & Strategic Recommendations

NewtonSchool ofTechnology | DataAnalytics CapstoneProject

TOPIC - Olist E-commerce Data Analysis

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Executive Summary

Project Context

Brazilian e-commerce platform Olist processed 100,000+ orders across 2016–2018. Analysis reveals delivery inefficiencies and customer satisfaction gaps concentrated in northern states, with logistics performance being the primary driver of review scores.

Strategic Objective

Identify root causes of late deliveries and low satisfaction scores using Python-based ETL and statistical analysis across 9 relational tables, and deliver data-backed recommendations to improve retention.

100K+ Orders

Comprehensivedataset covering 2016-2018

9 Tables

Orders,customers, sellers, payments, reviews

5 KPIs

Revenue,AOV, delivery, satisfaction, retention

Data Engineering Pipeline

Data Source

Kaggle Olist dataset with 100K+ orders and relational structure

Data Cleaning

Missing value imputation, date formatting, consistency checks

Integration

Table merging creating unified analytical dataset

Validation

Quality assurance and data dictionary documentation

Key Fields Analyzed

- Order ID & customer ID linkage
- Payment value and method tracking
- Review scores (1-5 scale)
- Delivery timestamps and regions
- Product categories and pricing
- Seller location and performance

Key Performance Indicators

Strategic metrics driving business growth and operational excellence

Total Revenue

Cumulative sales value indicating overall business performance and growth trajectory

Average Order Value

Mean transaction value guiding pricing strategy and upselling opportunities

Delivery Time

Days from order to delivery measuring logistics efficiency and customer wait time

Customer Satisfaction

Review scores (1-5) reflecting purchase experience and platform trust

Repeat Purchase Rate

Customer retention percentage indicating loyalty and long-term value creation

Exploratory Data Analysis Insights

Seasonal Patterns

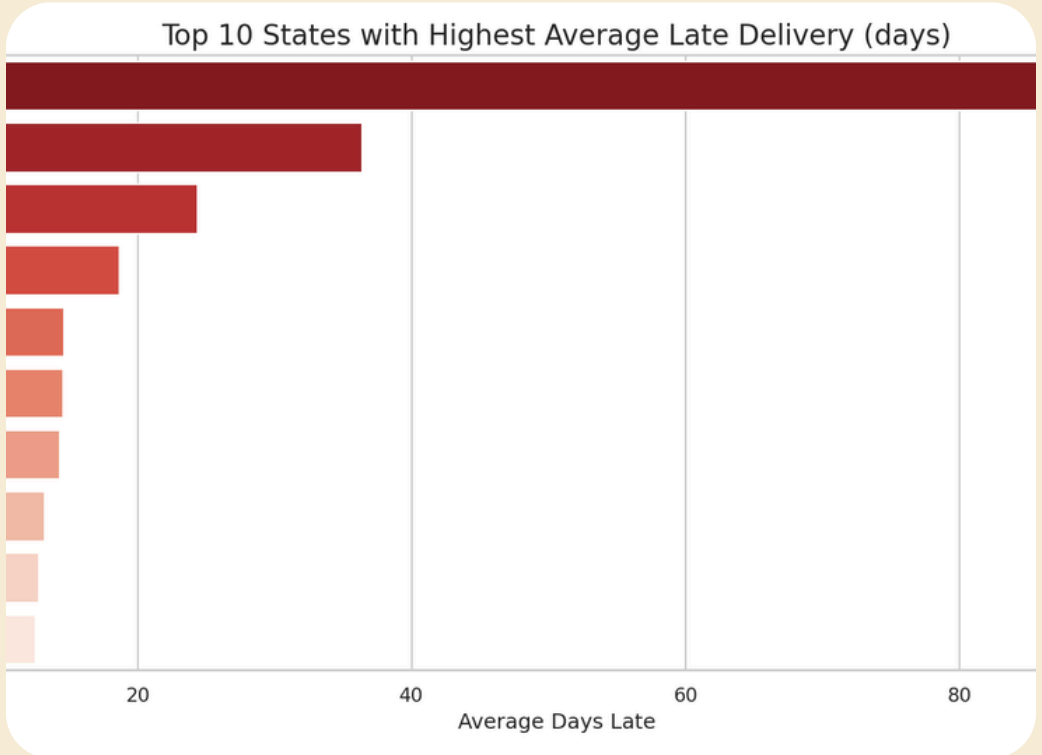
Order volume grew from near zero in mid-2016 to a peak of 8,800 orders in November 2017, a 40x increase. A dip in December 2017 suggests post-peak normalisation rather than decline.

Category Concentration

Top 3 categories by order volume are bed_bath_table (~11,000 orders), health_beauty (~9,600 orders), and sports_leisure (~8,600 orders). These three drive the majority of platform transactions.

Geographic Performance

SNorthern states show the worst late delivery performance. Amapá (AP) averages 95 days late when orders are delayed, followed by Roraima (RR) at 35 days and Amazonas (AM) at 25 days.



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Average Delivery Days

Overall logistics performance metric

4.0

Average Review Score

Customer satisfaction baseline

68%

Revenue Concentration

Top categories share

35%

Holiday Spike

Q4 revenue increase

Advanced Analytical Findings

Delivery-Time Correlation

Negativecorrelation(r = -0.62)between delivery days and review scores. Each additional day reduces satisfaction by 0.3 points

Customer Segmentation

High-valuecustomers(top 20%)generate 55% of revenue with 2.5x repeat purchase rate versus low-value segment

Payment Method Analysis

Credit cardusers (58%oforders)show 18% higher AOV and 23% better satisfaction vs boleto payments

Root Cause Identification

Logistics delays inremote regions (North/Northeast) cause 70% of negative reviews and 45% of order cancellations

Interactive Dashboard Architecture

Executive View

- Real-time KPI summary cards
- Revenue trend line charts
- Performance scorecards

Operational View

- Category performance matrices
- Regional delivery heatmaps
- Seller rating distributions

01

Date Range Selection

Filterbymonth,quarter,orcustom period

02

Category & Region

Drill down to specific segments

03

Payment Type

Analyzebypayment method

04

Order-Level Detail

Full transaction breakdown

Top Insights

Delivery delays hurt satisfaction

Directly reducing how customers feel about the experience.

Few categories drive revenue

A small number of categories contribute most of the total.

Some regions lag in logistics

Performance is uneven, with certain regions consistently underperforming.

Repeat customers boost revenue

Returning buyers significantly increase overall revenue.

Strategic Recommendations

Optimize Logistics Network

Invest in regional distribution centers in North/Northeast to reduce delivery times by 30%. Partner with local courier services for last-mile efficiency

Promote High-Value Categories

Allocate 60% marketing budget to Electronics, Fashion, and Home categories. Implement cross-selling algorithms to increase AOV by 25%

Launch Loyalty Program

Target top 20% customers with exclusive offers increasing repeat purchase rate from 18% to 32%. Expected revenue lift: $450K annually

Enhance Seller Analytics

Implement real-time seller scorecards tracking delivery time, review scores, and order volume. Flag underperformers for coaching or contract review

Expected Business Impact

Improved Customer Satisfaction

Reduce delivery time from 14.2 to 10 days, increasing average review scores from 4.2 to 4.6

Enhanced Operational Efficiency

Streamlined logistics reducing late deliveries by 45% and order cancellations by 30%

Revenue Growth from Retention

Boost repeat purchase rate from 18% to 32%, driving $450K+ incremental annual revenue

Next Steps & Conclusion

Immediate Actions

- Integratereal-timedashboards with operational systems
- Launch pilot logistics program in North region
- Design loyaltyrewards program framework

Future Enhancements

- Apply machine learningfor demand forecasting
- Expand supply chain analytics integration
- Scale analysis to multi-market expansion

Project Limitations

DatasetrestrictedtoBrazil (2016-2018), no real-time data integration, incomplete records requiring imputation, and assumptions in causal analysis

Next Steps

Forward-looking priorities to extend the project and turn insights into action.

Integrate real-time dashboards

Apply machine learning for demand prediction

Improve supply chain analytics

Expand to multi-market analysis

Thank You